It is named so because it can be adapted for a very wide range of milling operations.

Various milling attachments like index head, vertical milling head, slot milling head and rotary table can be mounted. It can be machine drills reamers, gears, milling cutters with a very high degree of accuracy and so it finds an important place in workshop.

c. Omniversal milling machine

In addition to the table movement obtained in a universal milling machine, the knee can be fited to a required angle. It is useful for machining helical grooves, reamer and bevel gears. It is mostly used in tool room work.

d. Vertical milling machine

A spindle of a vertical milling machine is positioned at right angles to the table. The cutter is moved vertically or at an angle by swiveling the vertical head of the machine.

The machine is adapted for machining slots and flat surfaces by moving the table. By mounting end mills and face milling cutters on the spindle, vertical milling and internal milling are preformed.

5.5 Differences between a plain milling machine and a universal milling machine

S.No.	Plain milling machine	Universal milling machine
1.	The table can be moved vertically, longitudinally and crosswise.	Apart from the three movements of a plain milling machine. It can be swiveled about 45^0.
2.	Helical milling works cannot be done as the table cannot be swiveled.	The table can be swiveled and helical milling and spiral milling can be performed.
3.	As there are no special attachments, operations like gear cutting, slotting and vertical milling operation cannot be performed.	Special attachments like indexing head, rotary table, vertical milling attachment, slotting head are available with this machine.
4.	It is more rigid and suitable for machining on heavy and large workpiece and for simple milling operations.	It is meant for light workpieces, a wide range of operations can be performed in this machine. It is mainly used in tool rooms.
5.	The cost is less.	It is very costly.

5.6 size of a milling machine

The size of a milling machine is specified as follows

1. The size of the table (Length and width)
2. The maximum lengths of longitudinal, cross and vertical travel of the table.

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